Comparative Analysis of Forman-Ricci and Ollivier-Ricci curvature on hypernetworks

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1 Introduction

Hypernetworks are rich data structures, capturing higher order relationships within the data. The use of higher order interactions than pairwise for modelling social and biological systems is naturally intuitive, for example communication between three or more people. As hypernetworks are more complex data structures than standard networks, performing analysis suffers from the drawback of being even more challenging than on networks.

Curvature can provide insightful topological information about a space; however, application to discrete data inherently requires an approximation to be made. Despite this, computation on networks is well established, with Ollivier-Ricci and Forman-Ricci curvature offer competing approaches to discretisation. [Ollivier (2007),Ollivier (2009),Forman (2003)] These two techniques come from different theoretical derivations and exhibit similarly distinct empirical properties, therefore investigating their dissimilarities is far from trivial.

Beyond geometric intuition, discrete curvature has successfully been applied to both shallow and deep machine learning tasks. [Tian et al. (2025),Fesser & Weber (2024)] Many of these approaches leverage Ricci flow, which gained prominence following Perelman's work in 2003 [Perelman (2003*a*), Perelman (2003*b*)], after which uses in statistics followed. For the continual development of statistical applications, understanding the choice between these two discretisations and the subsequent trade offs has been essential.

Despite the broad literature and ongoing research utilising discrete curvature on networks, extending this framework to hypernetworks has recieved considerably less attention, due to challenges with topological intuitiveness and computation. Despite the existence of theoretical techniques for calculating Ollivier-Ricci and Forman-Ricci curvature on hypernetworks [Saucan & Weber (2018),Yadav et al. (2022)], there is still exists a notable gap in the literature of a comparison between the two discretisations for hypernetworks on data. Furthermore, this empirical assessment is vital to understanding the statistical properties of discrete curvature, especially if its calculation precedes downstream machine learning tasks.

2 Literature Review

For discrete curvature on networks, Ollivier-Ricci and Forman-Ricci curvature have established themselves as the two leading approaches. [Ollivier (2007),Ollivier (2009),Forman (2003)]Fundamentally, Ollivier-Ricci curvature provides an elegant but computationally expensive discretisation, as it is derived from optimal transport. On the contrary, Forman-Ricci curvature has a more complex derivation; however, yields a more straightforward formula which can computed locally, allowing for faster computation. A comparison for networks has been provided by "Comparative analysis of two discretizations of Ricci curvature for complex networks" [Samal et al. (2018)], which lays out theoretical justification for the discrepancies in the data. The paper advocates for augmented Forman-Ricci curvature in certain practical situation, due to greater numerical similarity to Ollivier-Ricci than standard Forman-Ricci while maintaining fast computation. Augmented Forman-Ricci curvature includes "triangles" (when nodes x and y share another node in their neighbourhoods) in addition to the standard Forman-Ricci curvature, which contribute to positive curvature.

Regarding statistical applications of discrete curvature, clustering and coarsening have been prominent areas of study. [Tian et al. (2025), Weber et al. (2017)] Identifying clusters is accomplished by techniques which remove low curvature edges, which represent sparsity of the network. Conversely, areas of high curvature intuitively reflect being dense and often corresponds to areas of greater interest (for example, in social networks). Naively, this removal can be done through a basic iterative algorithm with thresholds; however, Ricci flow based techniques allow the networks to be manipulated in a more nuanced way.

Furthermore, discrete curvature has been applied to machine learning (ML) problems, particularly with graph neural networks (GNNs). An example of this is to reduce "oversquashing" and "oversqueezing" of message-passing in GNNs. [Fesser & Weber (2024)] To reiterate, the relevance of analysing discrete curvature itself to these ML tasks is great: understanding an input feature is critical to rigorous ML development.

Regarding hypernetworks, their distinction from networks is that their hyper-edges can be of higher order than pairwise. Their applied study is not new: the paper "Learning with Hypergraphs: Clustering, Classification, and Embedding" [Zhou et al. (2006)] from 2006 has almost 2000 citations now. This paper provides a foundational resource from which I can explore comparative statistical properties of hypernetworks for Part 4 in my Methodology.

For calculation, "A poset-based approach to curvature of hypergraphs" [Yadav et al. (2022)] provides an explicit way to compute the Forman-Ricci curvature. For Ollivier-Ricci curvature on hypernetworks, there exists a theoretical approach from [Saucan & Weber (2018)] based on polyhedral complexes, but the computation is novel. There also exists an approach, Ollivier-Ricci Curvature for Hypergraphs: A Unified Framework (ORCHID) [Coupette et al. (2022)], based upon clique decomposition of hypernetworks into networks and leveraging random walk theory for optimal transport.

In recent applied literature, "Higher-Order Learning with Graph Neural Networks via Hypergraph Encodings" [Pellegrin et al. (2025)] uses discrete curvature on hypernetworks. It also begins its Limitations section stating "A key limitation of this study, and many of the related works, is a lack of benchmarks, consisting of true hypergraph-structured data." This paper showcases both the use of discrete curvature on hypernetworks and the gap in the literature this thesis can help fill.

3 Methodology

This comparative analysis will be the culmination of numerous stages, each requiring synthesising theoretical and computational resources. Throughout, maintaining a statistical critique will help to provide insights that improves downstream machine learning capabilities.

First, a diverse a set of hypernetworks on which we will perform our analysis needs to established (set of 10-15). As [Pellegrin et al. (2025)] mentioned earlier, finding data which is naturally parameterised as a hypernetwork is important and itself a valuable component of this research. By using both real and synthetically generated data, our analysis will aim to verify relationships between the hypernetwork structure and its discrete curvature. Additionally, the existing ORCHID package has a tool which provides an estimate of computation cost, which I can use to establish an upper bound to the feasible size of hypernetworks being used, given this is expected to be the most computationally intensive method. See Appendix A for the key relevant github libraries for each method.

Beyond establishing the data set, the stages of progress can be broken down into five clear components:

1. Calculate the Forman-Ricci curvature on hypernetworks set.

2. Calculate the Ollivier-Ricci curvature on hypernetworks set. This will require implementing the theoretical approach based upon polyhedral complexes, along with comparison to ORCHID.

3. Conduct comparison, assessing both the theoretical and empirical discrepancies between the two different discretisations' outputs.

4. Evaluate the relationship between discrete curvature and statistically interesting properties of hypernetworks (such as embeddedness, which will need computed).

5. Perform basic clustering using both discretisations and assess the difference in behaviour.

I expect each of these to take a relatively similar length of time, a couple of weeks to complete, with the Ollivier-Ricci curvature taking slightly longer.

Computing Ollivier-Ricci curvature is what I anticipate to be the most challenging component. First, clearly establishing the theoretical similarities and differences between ORCHID and the polyhedral complexes approach will be allow this thesis to engage with the current literature. For the novel approach via polyhedral complexes, the implementation required is to construct the dual of hypernetwork. With the dual's nodes and edges constructed, we can apply utilise existing packages. I will conclude my findings on the relation between these two frameworks before proceeding with the following stages.

Regarding the comparative parts of the analysis, I intend to establish a novel and cohesive interpretation of the relationship between these approaches. The paper "Comparative analysis of two discretizations of Ricci curvature for complex networks" provides a guideline of how to conduct such analysis, combining intuitive and practical explanations. An obvious distinction is using statistical characteristics of hypernetworks; however, we identified in the literature review an primary resource which is long standing. For this comparison of Forman-Ricci and Ollivier-Ricci curvature, justifying the form(s) of correlation chosen will be necessary given the greater complexity in the edge structure, aiding interpretation of the results.

Finally, our analysis of clustering will aim to identify whether through pruning the edges of negative curvature, Forman-Ricci and Ollivier-Ricci curvature behave similarly to one another.

4 Conclusion

This thesis intends to provide a coherent and informative guide to the application of discrete curvature to hypernetworks. By examining the statistical relations between Forman-Ricci curvature, Ollivier-Ricci curvature and the hypernetwork itself, this work helps promote discrete curvature as viable tool for analysing hypernetworks.

By extension of the final part of the Methodology, this thesis demonstrates how it can provide a supporting role in developing more complex discrete curvature based ML techniques. Clustering and coarsening methods utilising Ricci flow have been extensively studied for networks, which follows directly from discrete curvature. Despite the increased difficulty of application to hypernetworks, approaches using Ricci flow on hypernetworks for has already been proposed [Sengupta et al. (2025)].

Furthermore, new literature offers hypernetworks as an alternative parametrisation to classical network problems in deep learning (Message Passing Hypergraph NNs Huang & Yang (2021), Transformers Liu et al. (2024)). For such techniques, discrete curvature may offer similar improvements for hypernetworks as existing research already offers on networks. [Fesser & Weber (2024)].

5 Appendix

Github libraries for computing Ollivier-Ricci and Forman-Ricci curvature:

1. ORCHID, Coupette et al. (2022), https://github.com/aidos-lab/orchid

2. Poset-Hypergraph-Curvature, Yadav et al. (2022), https://github.com/asamallab/Poset-Hypergraph-Curvature

3. GraphRicciCurvature for Ollivier-Ricci curvature on networks, Ni et al. (2019), https://github.com/saibalmars/GraphRicciCurvature

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